Christos Hadjinikolis

Engineering Manager | Energy Analytics & Production ML
Ph.D. Computer Science | UCL Research Associate | AI Standards Expert @ ISO/CEN-CENELEC JTC'21
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about me

Engineering Manager with 16 years across production ML, data platforms, consulting and AI research. At Vortexa, I lead engineering delivery for maritime intelligence supporting oil and gas analytics. I manage **6 direct reports** and lead a **10-person cross-functional team**, translating domain and product priorities into reliable data services and prediction systems. My work combines team development, stakeholder alignment, ML quality and platform strategy.

profiles
website
LinkedIn
GitHub
Google Scholar

leadership
6 direct reports
performance/career
delivery ownership
hiring/mentoring

energy analytics
maritime intelligence
destination/ETA
prediction quality
domain alignment

ML platforms
PyTorch
MLflow
model evaluation
data contracts
model-serving

engineering
Python/Java
Kotlin/SQL
Kafka/Flink
AWS/GCP
reliability

open source
PyPI: dynamicio
skeleton-replay
JetBrains plugin

work-experience

Vortexa Ltd, London, UK 12/2020–present — Engineering Manager | ML Systems Lead

I own engineering strategy and delivery for a live ML/data estate supporting maritime energy intelligence, turning about **6M** vessel-position records/hour into production intelligence for **13.5K** monitored vessels. Lead engineering, Data Science, Product and domain analysts on prediction quality, reliability and delivery.

- **People & Delivery:** Manage **6** direct reports spanning Product, SME analysis, Data Science, and Data Engineering; own performance/career development, team delivery, hiring for **6** roles, and delivery accountability for the wider team.
- **Product / Stakeholder Alignment:** Ran workshops across Product, SMEs, analysts, and engineers to turn model-quality disputes into shared definitions, measurable objectives, interface/SLA proposals, and OKR-linked workstreams.
- **Model Development & MLOps:** Led 0-to-1 research-to-production delivery for destination and arrival-time sequence/transformer models in **PyTorch**; established MLflow, model/data versioning, automated evaluation gates, replay, monitoring, and model-serving workflows at about **6M** events/hour.
- **Evaluation & Iteration:** Built batch/online model-evaluation and replay loops; analysed edge cases and failure modes with domain experts and Product to drive metric-guided model, data, and interface improvements.
- **Platform & Standardisation:** Standardised I/O seams, data contracts, repo archetypes, AWS CodeArtifact publishing, ADRs, dev containers, and local E2E tests (cut a **3h** DAG feedback loop to under **5** min).
- **Architecture & Reliability:** Led **Kafka Streams-to-Flink** as a strategic platform move; maintained signal-quality controls, rollback/fallback paths, shared on-call, and **MTTR** under **30** min.

DataReply, London, UK 04/2016–12/2020 — Senior Consultant | Data Scientist / ML Engineer

Joined Data Reply's London spin-off as its first consultant and helped grow it from **three practitioners and two managers to 30+ consultants**. Progressed to Senior Consultant; shaped opportunities, agreed deliverables/timelines, managed expectations, placed consultants, and delivered technical roadmaps across UBS, Vodafone, CNHi, and other enterprise clients.

- **Vodafone Group:** Led multi-workstream GCP data-science platform (based on Kubeflow) delivery; drove requirements, migration planning, CI/CD, workflow automation, ML lifecycle support, and feature-engineering migration.
- **CNHi:** Lead Data Scientist/Scrum Master on predictive maintenance (Azure/Databricks); translated stakeholder needs into DS/DE backlog and guided a **PySpark** team over live telemetry.
- **UBS:** Embedded consultant who grew from Data Scientist into ML Engineer through XP/pairing while delivering graph/process-mining analytics, observability, and real-time **Kafka**/Elasticsearch pipelines.

KCL, UCL, David Game College, UK 2010–2016 — Teaching and Technical Training

Delivered courses and tutorials in algorithm design, data structures, databases/SQL, and Java/Python during and after doctoral research; developed presentation discipline, learner support, and mentoring habits.

education

2010–2014 **Ph.D. in Computer Science** King's College London, UK

***Thesis**: Persuasion Dialogues & Opponent Modelling.*

Developed logical inference methods for large **Knowledge Graphs**, combining symbolic reasoning, **graph analytics**, and Bayesian techniques (e.g., MCMC). Received **Best Poster Award** at IJCAI 2013 (413 submissions), **EPSRC PhD Scholarship**, **Outstanding TA Award**, and **Graduate Certificate in Academic Practice**.

Access Publications on Google Scholar

2004–2010 **Diploma (BEng) in Computer Engineering** University of Thessaly, GR

Five-year polytechnic degree in Computer Science and Engineering with a **strong focus on mathematics** (10 courses), covering theoretical aspects, networking, and telecommunications. ***Majored in Artificial Intelligence***.

2002–2004 **Military Officer: Second Lieutenant** National Guard, CY

Completed national service as a commissioned Second Lieutenant.

selected engineering projects

Private **Promet** GenAI harness

Private applied AI project exploring voice, memory and tool-use workflows; hands-on work on schema validation, approval gates, durable state, traces and deterministic replay for auditable LLM interactions.

PyPI **skeleton-replay** Runtime evidence

Python developer tool that turns script/pytest runs into traces, architecture snapshots, workflow evidence, and replayable reports for review, debugging, onboarding, and LLM-assisted code understanding.

JetBrains **Skeleton Replay plugin** IDE workflow

Free PyCharm/IntelliJ plugin bringing Skeleton reports, workflow evidence, and source navigation into the IDE; reflects a bias for developer experience, evidence, and shared understanding.

PyPI **dynamicio** Data contracts

Free public tooling for making I/O seams explicit, validating schema/data expectations, and switching local/dev/prod datasets so pipeline logic can be tested with faster feedback.

committees/affiliations

since 10/2024 **University College London, Dept. of Information Studies** Associate Researcher

Research/teach practical AI standardisation: AI Act, auditability, model/data versioning, explainability, and safe adoption.

since 01/2021 **ISO - International Organisation for Standardisation** Committee Expert Member for JTC'21

Member of JTC'21 Working Group 3, developing AI standards aligned with EU policy and international norms.

talks/interviews

2023 **Bill Raymonds, Agile in Action** Podcast Interview

Agile Data Science: Sailing Through Data Science: The Agile Journey at Vortexa, **Details:** YouTube

2022 **Open Data Science Conference** Industry Conference Talk

dynamicio, a published **PyPI library** for abstracting I/O in ML/data workflows, **Details:** here

2020 **iunera: Big Data Innovators** Interview Blog Post

The Agile approach in data science explained by an ML expert, **Details:** here

2020 **Big Data Warsaw** Industry Conference Talk

Monitoring & Analysing Communication and Trade Events as Graphs, **Details:** here

2018 **Connected Data London** Panelist @ Industry Conference

AI tribes and interconnections: what's graph got to do with it?, **Details:** here

2018 **Minds Mastering Machines** Industry Conference Talk

A talk on doing Data Science the Agile Way, **Details:** here

certificates & awards

2017–2020 **Selected ML/cloud credentials** AWS, Google Cloud, O'Reilly

AWS ML Specialty; Google Professional Data Engineer; Apache Spark Developer.

2016–2019 **Data/graph engineering credentials** Coursera, Elastic, Neo4j

Process Mining; Graph Analytics; Advanced Elasticsearch; Neo4j Certified Professional.